Aadit Kshirsagar

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EDUCATION

University of Southern California Aug 2025 - May 2027
Master of Science in Computer Science *GPA:3.5/4*
Relevant Coursework*:* Foundations of AI, Machine Learning, Analysis of Algorithms, Natural Language Processing

Savitribai Phule Pune University Aug 2020 - Jun 2024
Bachelor of Engineering in Computer Science *GPA:3.61/4*
Relevant Coursework*:* Data Science, Cloud Computing, Deep Learning, Information Retrieval

TECHNICAL SKILLS

Programming: Python, Java, SQL, Git
Databases: PostgreSQL, MongoDB, Chroma, Pinecone, Milvus, Weaviate
Web Technologies: HTML, CSS, JavaScript, Streamlit
Cloud and DevOps: Azure DevOps, Google Cloud, Amazon Web Services, Fast API, Docker, Kubernetes, Azure AI, Open AI, Amazon Bedrock
AI/ML: Generative AI, NLP, LLMOps, MLOps, RAG, AutoGen, MCP, LangChain, LlamaIndex, LangGraph, Prompt Engineering, Computer Vison, Transformers, LoRa, Pandas, PySpark, PyTorch, TensorFlow, Keras

WORK EXPERIENCE

Calsoft Pvt Ltd | Data Science Intern | Pune, India Apr 2024 - Jun 2025
- Led end-to-end development of **scalable GenAI solutions** within the core team, from system design through optimization and deployment.
- Built multi-pipeline, agent-based RAG system, improving **test case relevance accuracy to 95%** and enabling retrieval of legacy test assets.
- Deployed an on-premises AI solution using open-source LLMs, **reducing manual test case generation by 80%** and significantly boosting QA productivity.

Intellect Design Arena Ltd | AI/ML Intern | Pune, India Sep 2023 - Mar 2024
- Engineered core components of IGTB Copilot, enabling Relationship Managers to extract actionable insights from bank data.
- Implemented LLM integration and prompt engineering to deliver a successful Copilot proof of concept, **automating payment onboarding workflows by over 70%.**

Intellect Design Arena Ltd | Software Developer Intern | Pune, India Jan 2023 - Mar 2023
- Engineered Natural Language Processing techniques to efficiently filter and structure bank customer data.
- Developed a Streamlit-based AI application that **improved KYC check efficiency by 45%.**

PROJECTS

Text Gen from Scratch | Codebase
- **Designed a 6-layer transformer** with custom positional encoding and learning rate scheduling to improve convergence and stability.
- Implemented a scalable end-to-end training pipeline for text generation with a **50k-word vocabulary and 64-token sequences, trained across 50 epochs.**

Medi Match Copilot | Codebase
- Integrated **multimodal Llama models** for clinical NLP and medical image analysis, enabling **combined text and vision reasoning with up to 95% diagnostic confidence.**
- Created a modular, scalable Python architecture supporting medical data retrieval, hypothesis generation, and visual analysis for long-term maintainability.
- Delivered an end-to-end clinical assistant automating symptom interpretation, imaging evaluation, and evidence-based knowledge synthesis to support diagnostic decision making.

Eco Policy Copilot | Codebase
- Supported evidence-based climate and economic policy analysis through **advanced data modeling** and automated workflows for environmental researchers.
- Built an AI-driven policy analysis platform using **LangChain** and **LangGraph**, integrating **6 data sources** and **20+ policy types** across climate, economic, and energy domains to generate actionable recommendations.
- Designed an **agentic architecture with autonomous agents** to ingest data, simulate policy scenarios, and produce executive summaries for efficient decision support.

RESEARCH AND PUBLICATIONS

- "**Enhancing RAG performance through Chunking and Text Splitting Techniques"** published in the International Journal of Scientific Research in Computer Science, Engineering, and Information Technology. | Link